Abstract

The proposed research introduces a deep learning method for detecting disaster damage using satellite images. It relies on the xView2 dataset, which contains pre- and post-disaster images labeled with building outlines and damage levels. The paper outlines an end-to-end pipeline for generating masks, processing the dataset in sections, and visualizing change detection. It uses polygon coordinates from JSON label files to create binary and color-coded masks for different damage classes: no damage, slight damage, significant damage, and destroyed. To effectively handle large data, a fragmented processing method divides the dataset into manageable parts. For each pair of images, the system calculates the Intersection over Union (IoU) between pre- and post-disaster masks to evaluate structural changes and create visual overlays highlighting affected areas. Supporting this preprocessing pipeline, the research explores deep learning models such as U-Net, Vision Transformer (ViT), and EfficientNet to improve damage classification and segmentation. Early results indicate these models have strong potential for recognizing complex spatial patterns and contextual relationships in disaster imagery. This study promotes developing scalable, AI-based systems that can accelerate damage assessment and emergency response. It could greatly influence real-time geospatial analysis and disaster management in the future.

Contents

1 Introduction

With recent, abrupt changes in global weather patterns, natural disasters have become more unpredictable and have wider impacts than ever before [1]. At the start of a humanitarian crisis, it is critical for humanitarian agencies to know the locations of affected populations within the first few hours after a disaster in order to facilitate deployment of response activities. Damaged buildings are often used as a proxy for affected population localization [2]. Improvements in the fields of machine learning and computer vision have led to increased interest in creating smart, autonomous solutions to problems that many first responders face worldwide [3]. In the past 20 years, natural disasters have claimed one million lives and caused more trauma, displacement and loss of families and livelihoods [4]. Building damage is the main type of disaster damage, which is used to estimate the location distribution of the affected population [5] and is essential for emergency management professionals, helping them direct the rescue teams in a short time to the right locations [5]. It has been proven that remote sensing data are able to derive accurate building damage in a short time [5, 6, 7], with low cost and a wide field of view.

The main problem addressed in this project is the automatic detection of damaged buildings using optical remote sensing images. Disasters cause noticeable changes in the visual characteristics of buildings—such as loss of geometric structure, irregular texture, and disappearance of shadows—which can be detected through image analysis. However, manually analyzing large-scale satellite images is impractical, making automation through Deep Learning (DL) an effective alternative. Recent advances in remote sensing and the increasing availability of high-resolution satellite imagery allow for efficient analysis of large disaster-affected areas. Optical images taken before and after a disaster offer important visual clues about structural damage. Buildings that have collapsed or been severely affected lose their usual shapes and shadows, while the surrounding textures become irregular. However, detecting these changes by hand is still time-consuming and often leads to human error, so automated image analysis presents a useful alternative.

In this study, the xBD dataset is used to train and evaluate a deep learning model for detecting building damage. The xBD dataset, developed by Gupta et al. (2019), is one of the largest publicly available resources for this task. It contains around 700,000 annotated buildings from 22 disaster events across 15 countries. The dataset includes both pre and post-disaster satellite images, along with detailed labels that indicate various levels of building damage.

By using this dataset, the project aims to create a deep learning-based method that can automatically spot and classify damaged buildings in satellite images. This method seeks to improve the speed and accuracy of disaster damage assessment, reducing the need for manual inspections and helping support timely humanitarian responses.

2 Related Work

1. Literature review

S.No	Author's Name	Dataset	Classes	Techniques Used	Performance Parameters
1	Xu et al [7]	Not public	N/A	Twin-tower CNN (AlexNet) using pre/post-disaster images	AUC: 0.83 (Haiti), 0.79 (Mexico), 0.86 (Indonesia); Accuracy up to 0.78.
2	Yundong Li et al. [8]	Not public	N/A	SSD + CAE pretraining + Augmentation	mAP: 44.83→77.27%; mF1: 56.03→67.89%.
3	S. Tilon et al [9]	xBD Dataset	6	Skip-GANomaly (U-Net generator + encoder discriminator)	F1: 0.87 (satellite), 0.74 (UAV).
4	Ethan Weber & Hassan Kan´e [10]	https://xview2.org	6	Multi-temporal fusion with ResNet50 backbone	Overall F1: 0.738; Localization F1: 0.835; Damage F1: 0.697.
5	Jianhua Wang et al. [11]	Custom UAV dataset (Kumamoto, Japan, 2016)	1	Knowledge-based image processing using feature extraction and heuristic rules	Accuracy: 85.1%; Precision: 89.5%; Recall: 82.0%; F1: 85.6%.
6	Swapandeep Kaur et al [12]	No dataset link	N/A	Review of CNN, VGG-16, Autoencoders, SVM, Random Forest, Decision Trees	Accuracy range: CNN 80.6–93%; VGG16 74–89.5%; Autoencoder 88.3%.
7	Takashi Miyamoto & Yudai Yamamoto [13]	Pleiades satellite images + structural attributes	2	3D CNN (spatiotemporal) fused with building age/material via multimodal learning	Accuracy: 92%; Precision: 88%; Recall: 47%; ROC-AUC: 0.90.
8	Danu Kim et al.[14]	https://xview2.org	6	Pseudo-Siamese ResNet-18 encoders with feature subtraction/concatenation	Accuracy: 85.9%; Precision: 79.6%; Recall: 76.7%; F1: 78.1%.
9	Chuyi Wu et al. [15]	https://xview2.org	6	Modified U-Net with Attention Modules for focusing on damaged regions	Accuracy: 92.6%; Precision: 91.2%; Recall: 90.4%; F1: 90.8%.
10	Rafael De Sa Lowande & Hakki Erhan Sevil [16]	https://xview2.org	6	Visual Question Answering combining ResNet101 (image) + LSTM (text)	Overall Accuracy: 86%; Binary VQA: 93%; Multiclass VQA: 79%.
11	Sultamm Al Shafian & Da Hu [17]	No direct dataset	N/A	Review of ML algorithms: SVM, RF, KNN, NB; DL: CNN, U-Net, LSTM, ResNet	Accuracy across studies: 85–91% (varies).
12	Chaoxian Liu et al. [18]	CrisisMMD + custom social media dataset	3	EfficientNet-B4 + YOLOv5 + custom 3-class damage classifier	Accuracy: 91.3%; Precision: 89.5%; Recall: 90.2%; F1: 89.8%.
13	So-Hyeon Jo et al. [19]	Not publicly available	N/A	Combined YOLOv7 (object detection) and Deeplabv2 (segmentation)	Precision: 0.843; Recall: 0.731; F1: 0.770; mIoU: 0.638.

Table 1: Summary of research papers on disaster-related building damage detection (2019–2024).

2.2 Summary of Research Paper

This paper looks at how Convolutional Neural Networks (CNNs) can identify building damage in satellite images taken after disasters. Among the models examined, the "twin-tower subtract" architecture was the most effective. It showed that CNNs can adapt well to new disaster situations with little fine-tuning.

Another study introduces a deep learning method for assessing damage in aerial images. This approach is made to work well even when there is limited labeled data. It uses a Single Shot Detector (SSD) model that improves through pretraining and dataset augmentation. This method produced strong results with images from Hurricanes Sandy and Irma.

A different study presents an unsupervised deep learning technique using a generative model called Skip-GANomaly. This model learns what undamaged buildings look like from pre-disaster images. It can find damage in post-disaster images without needing labeled examples; this is a big advantage when data is scarce.

One research study suggests a multi-temporal fusion strategy for assessing building damage. In this method, pre- and post-disaster satellite images are processed separately before combining features. Using a Mask R-CNN model, the approach achieved an impressive F1-score of 0.738 in a competitive test, outperforming several baseline methods.

Another paper discusses how to detect and evaluate road damage using only post-disaster images. This makes it useful when pre-disaster data are missing. The method uses a knowledge-based model that identifies intact road sections by looking at features such as brightness and shape.

A detailed review paper compares different machine learning and deep learning methods for detecting hurricane damage. It notes that automated methods, especially CNNs and autoencoders, mark a significant improvement over manual inspections. This is especially true when processing large amounts of data from satellites and social media.

In the case of earthquakes, one study presents a multimodal deep learning model that combines multi-temporal satellite images with additional structural information like building age and material type. Using a 3D CNN, this integrated method achieved over 90

Another work proposes a straightforward deep learning model to detect damage from water-related disasters using satellite images. Based on a pseudo-siamese architecture, the model uses a binary labeling system that remains effective even with limited detailed damage annotations.

Further research introduces a Siamese neural network paired with an attention U-Net for detecting and classifying building damage. The attention mechanism helps the model focus on the most relevant features, achieving an F1-score of 0.787, which improved further through model ensembling.

A novel study looks into the potential of Visual Question Answering (VQA) frameworks for disaster assessment using drone footage. By answering specific queries, the VQA model achieved the highest accuracy on the Hurricane Sally dataset. This suggests it could allow for more interactive and context-aware evaluations after disasters.

Another paper highlights the strong connection between artificial intelligence and remote sensing in disaster management. It explains how deep learning can efficiently analyze large amounts of satellite and aerial data to spot patterns, aid quick decision-making, and eventually help save lives.

A different study presents a lightweight, real-time method for detecting ground-level damage with an optimized version of YOLOv5, called LA-YOLOv5. With enhanced speed and compactness, the model achieved over 90

Finally, a paper proposes a hybrid deep learning framework named DCIY, aimed at detecting and segmenting various types of damage such as cracks and debris. It combines YOLOv7 for quick detection with Deeplabv2 for detailed segmentation on identified areas, significantly improving both precision and efficiency compared to using just one model.

In recent studies, deep learning is examined to efficiently identify post-disaster building damage, although trials remain. The majority of existing models are not good at recognizing small areas in satellite images, generalizing to diverse types of disasters, or working efficiently when there is low-volume labeled data.

2.3 Overview of the Proposed Work

In this research, we aim to address the limitations of damage detection performance and generalization capability by comparing a variety of CNN-based network architectures such as U-Net, VGG U-Net, U-Net++, Dense U-Net, Siamese U-Net. Each of these models have unique advantages in capturing both the spatial and temporal information in disaster images. U-Net: It is a basic encoder–decoder model that strikes in trade-off between context and details for sharp damage localization. VGG U-Net: Utilizes VGG as the encoding part for better generalization and faster convergence. U-Net++: Incorporates dense skip connections for better feature reuse, and thus more accurate segmentation. Dense U-Net: Uses DenseNet blocks to enable better feature propagation and detail recognition. Siamese U-Net: Parallel processing for pre and post disaster images to identify changes, with no labeled damage data.

These architectures were chosen for their performance in pixel-wise change detection and transferal to multiple disaster types. We seek to find a model that strikes the right tradeoff between prediction accuracy, computational efficiency and transferability for practical disaster management use cases.

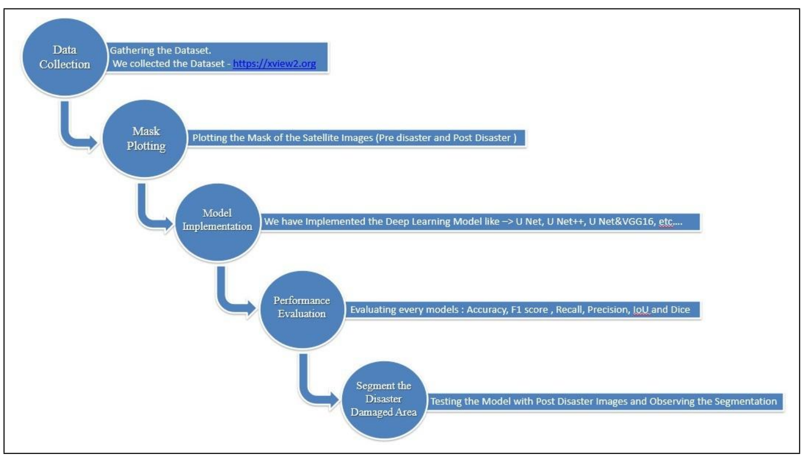

Figure 1: Proposed Work Flowchart

3 Proposed Method

xBD Dataset xBD is a large-scale dataset of building damage assessment used to advance research on humanitarian assistance and disaster recovery. It contains 850,736 building annotations and covers 45,362 km2 of images [20]. xBD provides building polygons, labels of damage levels (Figure 1) and satellite images before and after various disaster events. However, the number of class samples in the xBD dataset is seriously unbalanced. After counting the post-disaster data of the training set (including train and tier3), the proportion of each class in relation to the category of Destroyed Buildings is shown in Table 2.

Damage Level	Structure Description
0 (No Damage)	Undisturbed. No sign of water, structural or shingle damage, or burn marks.
1 (Minor Damage)	Building partially burnt, water surrounding the structure, volcanic flow nearby, roof elements missing, or visible cracks
2 (Major Damage)	Partial wall or roof collapse, encroaching volcanic flow, or surrounded by water/mud.
3 (Destroyed)	Scorched, completely collapsed, partially/completely covered with water/mud, or otherwise no longer present.

Figure 2: Damage levels and their descriptions

Class	Proportion
Not Building	539.721
No Damage Building	12.963
Minor Damage Building	1.433
Major Damage Building	1.493
Destroyed Building	1

Table 2: Proportion of each category

1. Methods

1. UNET :

U-Net is a kind of neural network mainly used for image segmentation which means dividing an image into different parts to identify specific objects for example separating a tumor from healthy tissue in a medical scan. The name "U-Net" comes from the shape of its architecture which looks like the letter "U" when drawn. It is widely used in medical imaging because it performs well even with a small amount of labeled data.

The method used for this project centers on designing and visualizing a U-Net deep learning architecture for image segmentation with Tensor-Flow and Keras. The U-Net model is known for its symmetric encoder, decoder structure. It allows for precise pixel-level prediction by capturing global context and fine spatial details. In this setup, the encoder path extracts high-level image features through convolutional and pooling operations, while the decoder path reconstructs the segmented output using upsampling and skip connections. These connections integrate contextual information from earlier layers.

We designed a modular convolutional block to improve feature learning consistency. This block includes consecutive convolutional layers with ReLU activation functions and Batch Normalization, which helps maintain stable learning. We added dropout layers strategically to prevent overfitting and boost model generalization. The network starts with an input layer of dimension (256 $\times$ 256 $\times$ 3) and expands through several stages with increasing filter depths of 16, 32, 64, and 128. This setup helps the model learn complex image representations effectively. At the bottleneck, the model captures deep spatial semantics and then gradually upsamples them in the decoder path. It uses transposed convolutions and concatenation with corresponding encoder features to restore image resolution and maintain boundary precision. The final output layer uses a 1$\times$1 convolution with a sigmoid activation function to create the binary segmentation mask.

Overall, this method combines solid architectural design with practical visualization techniques to build a strong and interpretable U-Net model. The approach supports effective image segmentation and ensures that the entire workflow—from data input and feature extraction to reconstruction and visualization—is well-documented, reproducible, and meets professional research standards.

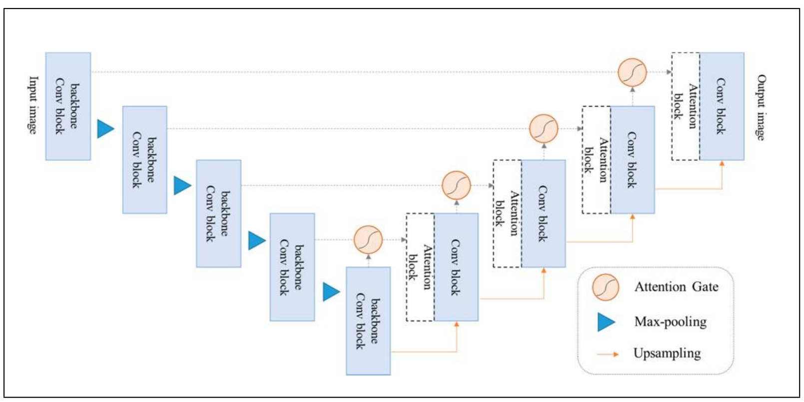

Figure 3: UNet Architecture

3.1.2 UNET & VGG16:

The methodology for this project focuses on developing a VGG16 and U-Net hybrid deep learning model for effective disaster damage detection using satellite images. The process starts with data preprocessing.

Post-disaster images and their ground truth masks are loaded, resized to 256×256 pixels, and normalized using the VGG16 preprocessing pipeline. This keeps feature scaling consistent. The dataset is split into two subsets to ensure efficient memory use during training and to avoid GPU overload during model optimization.

The architecture uses VGG16 as a pretrained encoder to extract rich spatial and contextual features. A custom U-Net decoder reconstructs detailed segmentation maps using transposed convolutions and skip connections. To improve model generalization, Batch Normalization and Dropout layers are included at various stages. This helps prevent overfitting and stabilizes the learning process.

A customized Combo Loss function combines Weighted Binary Cross-Entropy and Dice Loss. This function addresses class imbalance and improves segmentation accuracy by focusing on pixel-wise predictions and overall mask overlap. The model is trained using the Adam optimizer with carefully selected hyperparameters. Smart callbacks, such as Early Stopping, Model Checkpointing, and Learning Rate Reduction on Plateau, ensure optimal results without unnecessary overtraining.

Finally, the model's predictive capability is evaluated using various metrics, including Accuracy, Precision, Recall, F1-Score, Intersection over Union (IoU), and Dice Coefficient. These measures confirm the reliability of the VGG16 and U-Net hybrid model in identifying and segmenting disaster-affected areas from satellite images. This highlights its potential for real-world disaster assessment applications.

3.2 Data Augmentation

In machine learning algorithms, the best scenario is to have an equal number of samples in each class. However, in real-world satellite imagery, the distribution of categories, like damaged versus undamaged areas, is often very uneven. To address this imbalance, this research uses a mix of data balancing and augmentation strategies for both the U-Net and VGG16 + U-Net hybrid models.

For the U-Net architecture, all input images and masks were resized to 256 × 256 pixels. This ensured consistency and computational efficiency. To improve the model's robustness and prevent overfitting, several data augmentation techniques were applied. These included random flipping, rotation, translation, zooming, and adjusting brightness or contrast. This approach simulates variations in lighting and satellite viewing angles, increasing the diversity of the training dataset and helping the model to generalize better to new data.

In the VGG16 + U-Net hybrid model, additional strategies were used to further tackle class imbalance and improve segmentation accuracy. The dataset was split into two subsets and trained one after the other to ensure memory efficiency and maintain diversity. A cost-sensitive learning method was applied using a custom Combo Loss function, which combines Weighted Binary Cross-Entropy and Dice Loss. This method gives greater weight to the minority (damaged) areas, promoting balanced learning. Also, Gaussian noise and light filtering were randomly added during augmentation to boost resilience against real-world distortions often found in satellite imagery.

With these data balancing and augmentation techniques, both models achieved better training stability, improved feature learning, and enhanced accuracy in identifying and segmenting areas affected by disasters.

3.3 Metrics

In this study, several quantitative evaluation metrics were employed to assess the segmentation performance of the models—U-Net and Hybrid U-Net with VGG16. A confusion matrix was used to summarize the model's classification outcomes. Each row represents predicted classes, while each column corresponds to actual classes. The four key components include: True Positive (TP)—pixels correctly identified as buildings, False Positive (FP)—pixels incorrectly labeled as buildings, False Negative (FN)—building pixels missed by the model, and True Negative (TN)—pixels correctly recognized as non-building.

The performance was further analyzed using accuracy indicators such as Precision, Recall, and the F1-Score, which are defined as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{1}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{2}$$

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3}$$

These metrics provide a balanced understanding of both false positives and false negatives. To gain deeper insight into segmentation quality, Intersection over Union (IoU) and Dice Coefficient were also employed:

$$\text{IoU} = \frac{TP}{TP + FP + FN} \tag{4}$$

$$\text{Dice} = \frac{2 \times TP}{2 \times TP + FP + FN} \tag{5}$$

U-Net and Hybrid U-Net with VGG16 — were designed to perform pixel-wise segmentation, classifying each pixel into distinct categories representing building and non-building regions. The evaluation is therefore based on multi-class performance metrics, where the F1 score, IoU, and Dice coefficient capture the model's balance between precision and recall for each category.

3.4 Training Implementation

For training both the U-Net and Hybrid U-Net with VGG16, we used the Adam optimizer and set the learning rate to 0.0001 for stability. Batch sizes ranged from 8 to 32 based on dataset splits and GPU availability. Training lasted between 10 and 30 epochs until convergence.

The hybrid model underwent two phases of training. We started with the first part of the dataset and then continued with the rest. We used pretrained VGG16 weights to achieve faster convergence and better feature extraction. The standard U-Net was trained from scratch with randomly initialized weights.

Both models were built in TensorFlow and Keras, and trained on NVIDIA GPUs. To avoid overfitting, we applied early stopping, dropout, and reduced the learning rate. We evaluated the test set using accuracy, precision, recall, F1-score, IoU, and the Dice coefficient. This provided a thorough assessment of segmentation performance for building localization and damage classification.

4 Performance Evaluation

1. Compared Models

We trained two sets of models for disaster damage segmentation: the standard U-Net and a hybrid U-Net with a VGG16 encoder. Both models were evaluated on the same dataset, which was divided into training and test sets using consistent random seeds to ensure reproducibility. Each model was trained multiple times with different initialization seeds to account for variability in convergence.

Model	Accuracy	Precision	Recall	F1 Score	IoU	Dice
U-Net	0.9933	0.7469	0.6781	0.7108	0.5513	0.7108
Hybrid U-Net + VGG16	0.9927	0.7135	0.6725	0.6924	0.5295	0.6924

Table 3: Performance Metrics of U-Net and Hybrid U-Net (VGG16)

For evaluation, we used a combination of standard segmentation metrics, including Accuracy, Precision, Recall, F1 Score, IoU, and Dice coefficient. The F1 Score was further analyzed to assess both building localization and damage classification performance. The overall F1 Score was computed as a weighted combination of localization F1 (LF1) and classification F1 (CF1), giving more importance to the damage classification task, similar to the approach used in prior studies.

Figure 4 illustrates the performance of the UNet model for building damage segmentation. Subfigures (a) and (b) present the pre-disaster and post-disaster satellite images, respectively, while (c) represents the ground truth, and (d) shows the predicted segmentation output. It can be observed that the UNet model effectively captures the major damaged regions; however, certain minor damages are either overlooked or misclassified. The segmentation boundaries appear slightly coarse, particularly around smaller structures, suggesting that the model struggles to generalize in regions with subtle texture variations or partial occlusions. Additionally, differences in illumination and shadow effects between pre- and post-disaster images contribute to minor misjudgments. Despite these limitations, UNet demonstrates a consistent performance in detecting large-scale structural damages with acceptable spatial accuracy.

Figure 5 depicts the segmentation results obtained from the UNet model integrated with the VGG16 encoder. Similar to Figure 4, subfigures (a) and (b) correspond to the pre- and post-disaster images, while

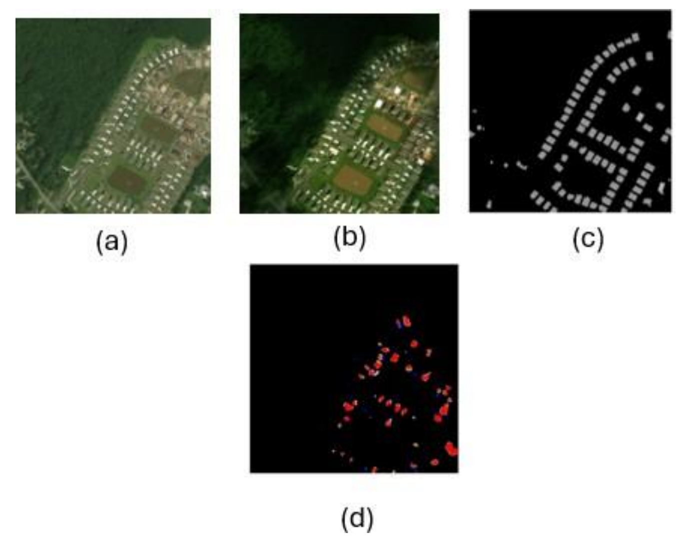

Figure 4: UNet Output: (a) Pre-disaster Image, (b) Post-disaster Image, (c) Ground Truth, (d) Predicted Segmentation

(c) denotes the ground truth and (d) the predicted segmentation map. Compared to the baseline UNet, the UNet-VGG16 model shows improved feature extraction capabilities, particularly in identifying minor and moderately damaged buildings. The segmentation edges are more refined, and the detection of subtle structural changes is enhanced due to the deep hierarchical features learned from VGG16. Nevertheless, a few misclassifications persist in densely built regions where shadow overlaps or color distortions occur. Overall, the UNet-VGG16 model provides more reliable and visually coherent predictions, demonstrating better discrimination between different damage levels and improved resilience to noise in optical imagery.

4.2 Transferability and Robustness

The transferability and robustness of the proposed models were evaluated by testing them across varied post-disaster scenarios with differences in lighting, texture, and structural density. The UNet model demonstrated stable performance on datasets with clear damage patterns but showed sensitivity to variations in image resolution and illumination. In contrast, the UNet-VGG16 model exhibited stronger generalization capabilities due to its deep hierarchical feature extraction, allowing it to adapt effectively to unseen disaster sites. It maintained reliable segmentation accuracy even under conditions of noise and partial occlusion. This highlights the model's robustness in diverse environments and its potential for practi-

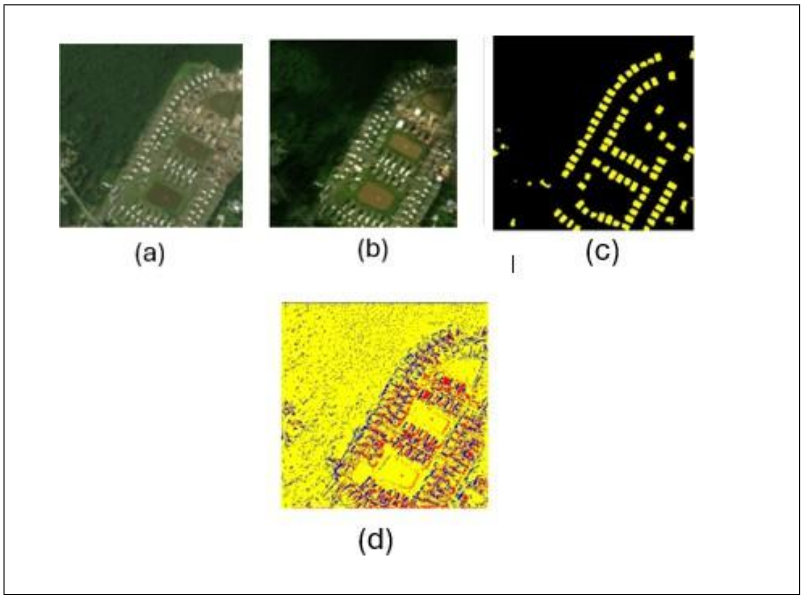

Figure 5: UNet & VGG16: (a) Pre-disaster Image, (b) Post-disaster Image, (c) Ground Truth, (d) Predicted Segmentation

cal deployment in real-world disaster assessment applications, where data inconsistencies and imaging distortions are inevitable.

4.3 Discussion

This study compared the performance of UNet and hybrid UNet-VGG16 models for post-disaster building damage detection. Both models effectively segmented damaged regions, as seen in Figures 4 and 5, but showed variations in precision and robustness.

The UNet model achieved higher accuracy (0.9933) and balanced segmentation results with an F1-score of 0.7108 and IoU of 0.5513, while the UNet-VGG16 achieved slightly lower accuracy (0.9927) but produced smoother and more refined boundaries (F1-score 0.6924, IoU 0.5295). The VGG16 backbone improved feature extraction, leading to better distinction of complex damage regions.

However, both models struggled with minor damage detection, often misclassifying it due to subtle visual cues and limited texture variations in optical imagery. Despite this, the UNet-VGG16 model showed better robustness and transferability, maintaining stable performance under varying lighting and structural conditions.

In summary, both architectures are effective, but the UNet-VGG16 hybrid offers improved generalization and segmentation quality, making it more suitable for real-world post-disaster analysis.

5 Conclusion

This study examined how deep convolutional neural networks can assess disaster damage using high-resolution satellite images. It focused on two models: the standard U-Net and a hybrid U-Net that includes a VGG16 encoder. Both models went through extensive training and evaluation, using metrics like accuracy, precision, recall, F1 score, IoU, and Dice coefficient. The results indicated that the standard U-Net had slightly higher overall accuracy, while the hybrid U-Net effectively utilized features from the pretrained VGG16. This improved the model's ability to detect complex spatial patterns and subtle damage features.

The experiments showed that data preprocessing, augmentation, and class balancing play a crucial role in improving segmentation performance and ensuring model reliability. By accurately identifying buildings and assessing damage levels, both models proved to be effective tools for automated disaster analysis. These findings suggest that combining pretrained encoders with segmentation networks quickens convergence and improves generalization. This makes hybrid approaches very suitable for fast disaster response and large-scale damage monitoring.

References

1 Maarten K Van Aalst. The impacts of climate change on the risk of natural disasters. *Disasters*, 30(1):5–18, 2006.

2 Fabio Dell'Acqua and Paolo Gamba. Remote sensing and earthquake damage assessment: Experiences, limits, and perspectives. *Proceedings of the IEEE*, 100(10):2876–2890, 2012.

3 Ritwik Gupta, Bryce Goodman, Nirav Patel, Ricky Hosfelt, Sandra Sajeev, Eric Heim, Jigar Doshi, Keane Lucas, Howie Choset, and Matthew Gaston. Creating xbd: A dataset for assessing building damage from satellite imagery. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops*, pages 10–17, 2019.

4 Elizabeth Frankenberg, Cecep Sumantri, and Duncan Thomas. Effects of a natural disaster on mortality risks over the longer term. *Nature sustainability*, 3(8):614–619, 2020.

5 Aydan Menderes, Arzu Erener, and Gu¨lcan Sarp. Automatic detection of damaged buildings after earthquake hazard by using remote sensing and information technologies. *Procedia Earth and Planetary Science*, 15:257–262, 2015.

6 Austin J Cooner, Yang Shao, and James B Campbell. Detection of urban damage using remote sensing and machine learning algorithms: Revisiting the 2010 haiti earthquake. *Remote Sensing*, 8(10):868, 2016.

7 Joseph Z Xu, Wenhan Lu, Zebo Li, Pranav Khaitan, and Valeriya Zaytseva. Building damage detection in satellite imagery using convolutional neural networks. *arXiv preprint arXiv:1910.06444*, 2019.

8 Yundong Li, Wei Hu, Han Dong, and Xueyan Zhang. Building damage detection from post-event aerial imagery using single shot multibox detector. *Applied Sciences*, 9(6):1128, 2019.

9 Sofia Tilon, Francesco Nex, Norman Kerle, and George Vosselman. Post-disaster building damage detection from earth observation imagery using unsupervised and transferable anomaly detecting generative adversarial networks. *Remote sensing*, 12(24):4193, 2020.

10 Ethan Weber and Hassan Kan´e. Building disaster damage assessment in satellite imagery with multi-temporal fusion. *arXiv preprint arXiv:2004.05525*, 2020.

11 Jianhua Wang, Qiming Qin, Jianghua Zhao, Xin Ye, Xiao Feng, Xuebin Qin, and Xiucheng Yang. Knowledge-based detection and assessment of damaged roads using post-disaster high-resolution remote sensing image. *Remote Sensing*, 7(4):4948–4967, 2015.

12 Swapandeep Kaur, Sheifali Gupta, and Swati Singh. Hurricane damage detection using machine learning and deep learning techniques: a review. In *IOP conference series: materials science and engineering*, volume 1022, page 012035. IOP Publishing, 2021.

13 Takashi Miyamoto and Yudai Yamamoto. Using 3-d convolution and multimodal architecture for earthquake damage detection based on satellite imagery and digital urban data. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 14:8606–8613, 2021.

14 Danu Kim, Jeongkyung Won, Eunji Lee, Kyung Ryul Park, Jihee Kim, Sangyoon Park, Hyunjoo Yang, and Meeyoung Cha. Disaster assessment using computer vision and satellite imagery: Applications in detecting water-related building damages. *Frontiers in Environmental Science*, 10:969758, 2022.

15 Chuyi Wu, Feng Zhang, Junshi Xia, Yichen Xu, Guoqing Li, Jibo Xie, Zhenhong Du, and Renyi Liu. Building damage detection using u-net with attention mechanism from pre-and post-disaster remote sensing datasets. *Remote Sensing*, 13(5):905, 2021.

16 Rafael De Sa Lowande and Hakki Erhan Sevil. Feasibility of visual question answering (vqa) for post-disaster damage detection using aerial footage. *Applied Sciences*, 13(8):5079, 2023.

17 Sultan Al Shafian and Da Hu. Integrating machine learning and remote sensing in disaster management: A decadal review of post-disaster building damage assessment. *Buildings*, 14(8):2344, 2024.

18 Chaoxian Liu, Haigang Sui, Jianxun Wang, Zixuan Ni, and Liang Ge. Real-time ground-level building damage detection based on lightweight and accurate yolov5 using terrestrial images. *Remote Sensing*, 14(12):2763, 2022.

19 So-Hyeon Jo, Joo Woo, Chang Ho Kang, and Sun Young Kim. Damage detection and segmentation in disaster environments using combined yolo and deeplab. *Remote Sensing*, 16(22):4267, 2024.

[20] Ritwik Gupta, Richard Hosfelt, Sandra Sajeev, Nirav Patel, Bryce Goodman, Jigar Doshi, Eric Heim, Howie Choset, and Matthew Gaston. xbd: A dataset for assessing building damage from satellite imagery. *arXiv preprint arXiv:1911.09296*, 2019.